

Adaptive Semi-Supervised Learning-Based Misbehavior Detection in Vehicular Networks

Elnaz Limouchi , *Member, IEEE*, and Francois Chan , *Senior Member, IEEE*

Abstract—Intelligent Transport Systems (ITS) use advanced technologies to enhance safety, mobility, and traffic efficiency for drivers, passengers, and pedestrians. Vehicular Ad-hoc Networks (VANETs) are a core component of ITS, enabling real-time communication among vehicles. However, ensuring VANET security remains a critical challenge, as malicious behavior can directly impact passenger safety and system reliability. Recent studies have explored machine learning (ML)-based misbehavior detection, predominantly relying on supervised approaches that require large volumes of labeled data, which are costly and difficult to obtain in practice. Semi-supervised learning provides a practical alternative by exploiting both labeled and unlabeled data. In this work, an adaptive semi-supervised self-training framework for VANET misbehavior detection is proposed. The method incorporates an adaptive confidence thresholding mechanism and hyperparameter optimization to improve learning stability under varying data conditions. Evaluation on the VeReMi dataset demonstrates that the proposed approach, particularly with XGBoost, consistently improves accuracy, precision, recall, and F1-score across different proportions of initially unlabeled data. The results further indicate robust performance in low-label scenarios, with the most notable gains observed at moderate-to-high proportion of unlabeled data. These findings highlight the robustness and practical applicability of the proposed method for real-world vehicular security applications.

Index Terms—VANET, semi-supervised learning, adaptive learning, self-training, misbehavior detection, network security, intrusion detection system.

I. INTRODUCTION

IN MODERN transportation, vehicular communication networks play a key role by enabling vehicles to share data instantaneously. This supports advanced safety features such as collision avoidance, helping to reduce the number and impact of accidents. As connected and autonomous vehicles develop, these networks will be crucial in making transportation safer, more efficient, and convenient, while also helping to reduce traffic congestion and environmental problems [1].

Detecting misbehavior in vehicular communication networks requires identifying and mitigating malicious activities

displayed by vehicles or entities within the network, encompassing threats such as DoS attacks, Sybil attacks, Black Hole attacks, Gray Hole attacks, and routing misbehavior. Ensuring network integrity, reliable communication, and the safety of vehicles and passengers is crucial. However, the dynamic and decentralized nature of vehicular networks presents challenges that include bandwidth restrictions, high mobility, and unreliable communication links. Developing robust and customized misbehavior detection mechanisms is essential to maintain network security and reliability.

Semi-supervised learning employs a combination of labeled and unlabeled data for model training, serving as a bridge between supervised and unsupervised learning methodologies. It provides advantages in scenarios with limited or expensive labeled data, striving to improve model performance by using labeled and unlabeled samples [2].

Self-training is a widely employed technique in semi-supervised learning methodologies. Initially, a model is trained using a limited set of labeled data. Subsequently, this model is used to classify unlabeled data, with instances having high confidence predictions added to the labeled dataset. The model is then retrained using this augmented labeled dataset, repeating the process until convergence or a predefined stopping criterion is reached. Self-training utilizes the model's own predictions to iteratively improve its performance on unlabeled data, thus improving its overall efficacy in semi-supervised learning tasks [3].

In vehicular networks, acquiring labeled data becomes challenging and expensive with network expansion. However, semi-supervised learning optimizes resource utilization by effectively utilizing both labeled and unlabeled data, which results in enhancing scalability in misbehavior detection systems. By integrating unlabeled data, these techniques adapt detection mechanisms to the dynamic conditions of vehicular networks, ensuring efficient misbehavior detection without the need for extensive manual labeling.

Limited research has been conducted on semi-supervised learning-based misbehavior detection systems proposed for vehicular networks. Most existing approaches rely entirely on supervised learning methods, which require extensive labeled data and often fail to generalize to unseen attack patterns. Hence, we were motivated to develop a semi-supervised misbehavior detection system in the context of vehicular networks. This system uses labeled and unlabeled data to reduce dependency and effectively detect misbehavior within the network.

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(Corresponding author: Elnaz Limouchi.)

The authors are with the Department of Electrical and Computer Engineering, Royal Military College of Canada, Kingston, ON K7K 7B4, Canada (e-mail: elnaz.limouchi@ieee.org; chan-f@rmc.ca).

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TABLE I
SUMMARY OF RELATED WORK ON SEMI-SUPERVISED LEARNING APPROACHES FOR MISBEHAVIOR DETECTION IN VARIOUS NETWORK ENVIRONMENTS

Method	Core Characteristics	Datasets	Reference
ASVM + FCM Hybrid IDS	Active Learning SVM with Fuzzy Clustering for low-label training	NSL-KDD	[4]
SVM + DBN Semi-supervised IDS	Combines SVM classification with Deep Belief feature learning	KDD 1999	[5]
Semi-Supervised Deep Learning for Intrusion Detection (SS-Deep-ID)	Hierarchical Semi-Supervised Training Approach, Integration with Fog-Enabled IoT Networks	CIC-IDS2017, CIC-IDS2018	[6]
Semi-supervised Federated Learning	Integration of Federated Learning with Semi-Supervised Learning	UNSW-NB15	[7]
Semi-supervised Adversarial Autoencoder (SAAE)	Integration of Generative Adversarial Networks (GANs) and Adversarial AutoEncoder (AAE)	NF-CSE-CIC-IDS2018, NF-UNSW-NB15	[8]
Double Mixup and Contrastive (2MiCo)	Data Augmentation Techniques at the input and hidden layers	Modbus Attack DataSet for AMI (MAMI)	[9]
Semi-supervised Fine-grained Intrusion Detection System (SF-IDS)	Fine-Grained Intrusion Detection with Self-Training and Pseudo-Label Filtering	NSL-KDD, CICIDS2017	[10]
DeepSemi-Supervised Anomaly Detection (Deep-SAD)	Federated Learning integration for Semi-Supervised Anomaly Detection	WUSTL-IIoT-2021	[11]
Autoencoder-based Anomaly Detection	Semi-supervised training and reconstruction error-based anomaly detection	NSL-KDD	[12]
Semi-supervised Vehicular Misbehavior Detection framework based on transformer (SVMDformer)	Semi-Supervised Transformer-Based Framework for converting vehicular message sequences into misbehavior scores and detecting misbehavior based on a threshold.	VeReMi-V2	[13]
Semi-supervised FL for V2X	SSL + FL to reduce labeling and bandwidth constraints	Extended VeReMi	[14]
FL for Corrupted BSM Detection	Privacy-preserving collaborative SSL learning	Vehicular BSM Data	[15]
Self-supervised NIDS for 5G-V2X	Pretraining on unlabeled data for privacy-preserving intrusion detection	CIC-IDS2017	[16]

The contributions of this study are summarized as follows:

- A centralized semi-supervised misbehavior detection framework for vehicular ad-hoc networks (VANETs) is proposed, using an adaptive self-training mechanism to effectively utilize large volumes of unlabeled data.
- An adaptive confidence thresholding strategy is introduced within the self-training process, where the threshold is iteratively updated based on model performance to improve the reliability of pseudo-label selection.
- A systematic analysis of the proposed threshold adaptation mechanism is conducted using the parameter "variation value", demonstrating its impact on the stability of the model and detection performance.
- The proposed framework is evaluated under varying proportions of initially unlabeled data, highlighting its robustness and effectiveness in realistic low-label scenarios typical of VANET environments.
- The approach is validated on the VeReMi dataset using representative classifiers (Logistic Regression and XG-Boost), showing consistent improvements over baseline semi-supervised configurations.

These contributions provide a scalable and data-efficient framework for misbehavior detection in vehicular networks and contribute to advancing semi-supervised learning methodologies in this field.

The remainder of this paper is structured as follows: Section II provides a brief overview of related work on semi-supervised learning-assisted misbehavior detection models in wireless networks. Section III outlines the methodology of this study. In Section IV, we present our proposed adaptive self-training misbehavior detection framework for vehicular ad-hoc networks. Section V presents the performance evaluation results. Finally, Section VI concludes the paper.

II. RELATED WORK

In this section, we review semi-supervised studies and methods that have been proposed for improving misbehavior detection in wireless networks. Table I provides a summary of these methods.

The authors in [4] propose an optimized, low-cost semi-supervised IDS that performs effectively with limited labeled data while supporting adaptive learning with minimal overhead. The proposed hybrid approach combines Active Learning Support Vector Machines (ASVM) with Fuzzy C-Means (FCM) clustering. The NSL-KDD dataset is used to train the model. Although the authors claim that the model is low-cost, the FCM + SVM approach can be computationally expensive for large datasets and requires careful parameter tuning to achieve effective performance.

In [5], a semi-supervised approach using Support Vector Machine and Deep Belief Network (DBN) methods for network intrusion detection is proposed, using a small amount of labeled data and a large amount of unlabeled data to classify the KDD 1999 dataset and detect specific attacks. This model is sensitive to parameter tuning, as its performance depends on the careful selection of parameters, such as SVM kernels or DBN layer configurations.

The authors in [6] present SS-Deep-ID, a semi-supervised framework for intrusion detection. SS-Deep-ID is designed for integration into fog-enabled IoT networks, delivering real-time intrusion detection. It includes a multiscale residual temporal convolutional (MS-Res) module to improve the network's capacity for learning spatiotemporal representations. A traffic attention (TA) mechanism is used to prioritize important information during the learning process. Additionally, a hierarchical semi-supervised training approach is introduced to use the sequential patterns in IoT traffic data. While effective for IoT

environments, this approach is computationally intensive and its applicability to highly dynamic vehicular environments remains unexplored.

A federated semi-supervised learning framework for Intrusion Detection Systems (IDS), incorporating network softwarization to streamline automation and deployment is presented in [7]. The method merges federated Learning with semi-supervised learning, where clients utilize unlabeled data to train unsupervised models for feature extraction, while the server trains a supervised model on labeled data. This method enhances privacy and scalability, however, it requires stable communication among distributed nodes, which may not be feasible in high-mobility vehicular networks.

The authors of [8] present a semi-supervised IDS model called the Semi-supervised Adversarial Autoencoder (SAAE), integrating Generative Adversarial Networks (GANs) and Adversarial AutoEncoder (AAE). Experimental validation is conducted on two widely used public datasets for benchmarking ML-based IDS, namely NF-CSE-CIC-IDS2018 and NF-UNSW-NB15. Although the adversarial setup enhances representation learning, the model's training complexity and dataset dependence limit its real-time deployment potential in vehicular contexts.

In [9], 2MiCo (Double Mixup and Contrastive), a novel method designed for semi-supervised learning in tabular security datasets, is proposed. These datasets present distinct challenges for machine learning, including contextual information loss and imbalanced data. To tackle these challenges, the proposed approach employs triplet mixup data augmentation at the input layer and common mixup in the hidden layer. Results from experiments are derived from binary and multi-class datasets found within the Modbus Attack DataSet for Advanced Metering Infrastructure (AMI). Despite improved performance on Modbus Attack DataSet for AMI (MAMI) datasets, the method is domain-specific and not optimized for vehicular communication data, which show different statistical properties.

The authors in [10] present SF-IDS, a novel semi-supervised fine-grained intrusion detection framework, aimed at addressing attack classification challenges in scenarios with limited labels and highly imbalanced classes. SF-IDS incorporates a self-training backbone model, RI-1DCNN, which enhances feature extraction by reconstructing input samples into a multichannel image format. The framework evaluates the uncertainty of generated pseudo-labels and employs this information, alongside prediction probability, for pseudo-label filtering. However, its static pseudo-label confidence threshold may lead to error propagation when the unlabeled data distribution shifts, an issue particularly relevant to highly dynamic VANETs.

The study in [11] explores a semi-supervised approach, termed "Deep-SAD," to address the challenge of partial data knowledge. To accommodate the large dataset size and privacy concerns, the model is integrated with a federated learning (FL) framework called "Flower," which distributes the learning process across five industrial sectors. The evaluation is conducted using the WUSTL-IIoT-2021 dataset, a simulation of an actual plant with injected threats. The study presents and assesses a framework for semi-supervised anomaly detection, beginning

with feature engineering. While effective for distributed industrial IoT, the approach assumes consistent data distributions across clients, which does not hold in vehicular networks with nonstationary and heterogeneous traffic patterns.

An autoencoder-based approach trained semi-supervisedly for anomaly detection in network traffic is introduced in [12]. The autoencoder learns from normal network traffic to derive a concise representation of regular data. Anomalies are identified based on reconstruction error. The solution emphasizes computational efficiency. Experiments are conducted on the NSL-KDD dataset. However, its reliance on low-dimensional feature representations limits its ability to capture complex or subtle misbehavior patterns, particularly in dynamic vehicular environments.

In the context of vehicular networks, the authors in [13] introduce SVMDFormer, a Semi-supervised Vehicular Misbehavior Detection framework based on Transformer. SVMDFormer converts vehicular message sequences into misbehavior scores and detects misbehavior by establishing a threshold. The framework is trained on a cloud server and deployed on an edge server to detect misbehavior from incoming vehicular messages in Internet of Vehicles (IoV). Although promising, SVMDFormer lacks an adaptive learning mechanism to refine thresholds based on confidence, and its reliance on the VeReMi dataset restricts generalizability due to the dataset's limited attack diversity.

The paper [14] proposes a semi-supervised federated learning-based misbehavior detection system for V2X communications to address the scarcity of labeled data and reduce bandwidth consumption. The approach is further applied to an extended version of the VeReMi dataset.

In [15], a semi-supervised federated learning framework is proposed to detect corrupted Basic Safety Messages in vehicular networks. The approach enables privacy-preserving, label-efficient misbehavior detection by allowing vehicles and a server to collaboratively train models without sharing raw data.

The study in [16] addresses emerging security challenges in 5G-enabled vehicular networks, where advanced technologies such as network slicing, SDN, and MEC introduce new vulnerabilities and privacy concerns. To overcome the limitations of data-hungry and privacy-sensitive intrusion detection systems, the paper proposes a privacy-preserving self-supervised learning-based NIDS for 5G-V2X networks. By pre-training models on large volumes of unlabeled data and fine-tuning them with a small, carefully labeled dataset, the approach enables effective intrusion detection while preserving sensitive vehicular information and enhancing protection against cyber attacks. The CIC-IDS2017 dataset is used to evaluate the proposed privacy-preserving self-supervised learning-based intrusion detection system for 5G-V2X networks.

Although prior studies have shown the potential of semi-supervised learning for intrusion detection, their applicability to vehicular networks remains limited due to several factors including reliance on domain-specific datasets, use of static confidence or pseudo-label thresholds, limited adaptability to dynamic and imbalanced environments, and computational inefficiency for real-time edge deployment. To overcome these limitations, this

study introduces an Adaptive Self-Training–based misbehavior detection framework that iteratively refines its confidence threshold to more effectively exploit unlabeled data.

Several recent studies have focused on anomaly and intrusion detection within intra-vehicle networks. The work in [17] introduces a self-supervised approach for in-vehicle anomaly detection using noised pseudo normal CAN traffic, while the study in [18] provides a comprehensive comparison of supervised and semi-supervised machine learning methods for CAN-based in-vehicle networks, proposing a semi-supervised deep autoencoder that outperforms other semi-supervised techniques. More recently, the authors in [19] propose a federated semi-supervised intrusion detection system that combines convolutional autoencoders with federated learning to reduce labeling and communication overhead in in-vehicle networks. Although these works show effective solutions for securing intra-vehicle communications, they focus on internal vehicle networks such as CAN and are therefore outside the scope of this paper, which instead addresses misbehavior detection in inter-vehicle communications.

III. METHODOLOGY

A. Semi-Supervised Learning

Semi-supervised learning utilizes both labeled and unlabeled data to improve classification accuracy. It can be particularly beneficial in domains with limited labeled data. Experts may manually label some data, but often struggle to label all instances, leading to a shortage of labeled data.

In semi-supervised learning, the pattern set S is split into two subsets:

- Labeled subset, $S_l = \{(x_i, y_i)\}_{i=1}^l$, where x represents the pattern, y is its associated label, and l is the number of labeled instances,
- Unlabeled subset, $S_u = \{(x_j)\}_{j=l+1}^{l+u}$, where x denotes the pattern and u is the number of unlabeled instances.

Typically, the number of unlabeled instances $|S_u|$ exceeds the number of labeled instances $|S_l|$.

Various algorithms, such as self-training, mix models, co-training, graph-based methods, and semi-supervised support vector machines, exist for handling partially labeled datasets. In this study, self-training was chosen as the primary approach due to its suitability for the research goals.

Self-training is one of the earliest methods for classifying unlabeled data using a smaller set of labeled data. It employs a wrapper algorithm, which utilizes a supervised approach for feature selection. In self-training, a supervised learning method is iteratively applied, initially trained only on labeled data. During each iteration, a portion of the unlabeled instances is assigned labels based on the current decision function. These newly labeled instances, along with their predicted labels, are then added to the training set. The classifier is retrained, and this process continues until all unlabeled data has been classified.

In [20], an enhanced version of the self-training algorithm was introduced, incorporating a confidence parameter as a threshold for incorporating new instances into the labeled dataset. Instances with prediction confidence equal to or greater than

the specified threshold (set at 0.6) are included in the labeled dataset.

The authors in [21] propose methods for dynamically calculating confidence rates to select labels in semi-supervised labeling, particularly in self-training and co-training frameworks. We adapt and revise the Flexible Confidence Classifier (FlexCon-C) approach presented in [21] to develop a misbehavior detection scheme tailored for vehicular communications. This scheme relies on adaptive self-training as its fundamental methodology.

IV. PROPOSED ADAPTIVE SELF-TRAINING MISBEHAVIOR DETECTION MODEL

A. Network Model

Our proposed system is managed by a central authority that handles vehicle registration and the issuance of keys for vehicular communication in VANET environments. Communication is limited to registered vehicles, and messages are authenticated using private keys.

In this network architecture, the system is trained once and then deployed on roadside units (RSUs), reducing hardware requirements due to offline training. The machine learning algorithm verifies message validity from vehicles to RSUs. Any detected issues prompt updates to legitimate vehicles, infrastructure, and the central authority, with misbehaving vehicles being revoked, which means that the misbehavior detection system is an active model.

B. Misbehavior Detection Model

In the context of misbehavior detection, the self-training technique is highly pertinent. This method includes training a model iteratively using a combination of labeled and unlabeled data. Initially, the model learns from the limited labeled data and then uses its predictions to label additional unlabeled data points. These newly labeled instances are subsequently incorporated into the training set for further refinement, gradually enhancing the model's efficacy. Given the scarcity and costliness of labeled data in intrusion detection systems (IDSs), which represent known attacks or benign behavior, self-training proves valuable by harnessing the ample unlabeled data available, such as network traffic or system logs. Through this iterative process, the misbehavior detection system can bolster its detection capabilities, expanding the labeled dataset and enhancing its capacity to identify potential threats.

Fig. 1 shows the overall workflow of the proposed self-training framework. As shown in Fig. 1, an initial classifier is trained using labeled data and subsequently applied to unlabeled samples to generate pseudo-labels. These pseudo-labeled instances are merged with the original labeled dataset to form an augmented training set, which is then used to re-train the classifier in successive iterations. The aim of the proposed model is to offer flexibility in determining the confidence threshold for incorporating new instances into the labeled dataset, aligning the labeling process with misbehavior detection scenarios. In this adaptive self-training method, the confidence threshold is dynamically

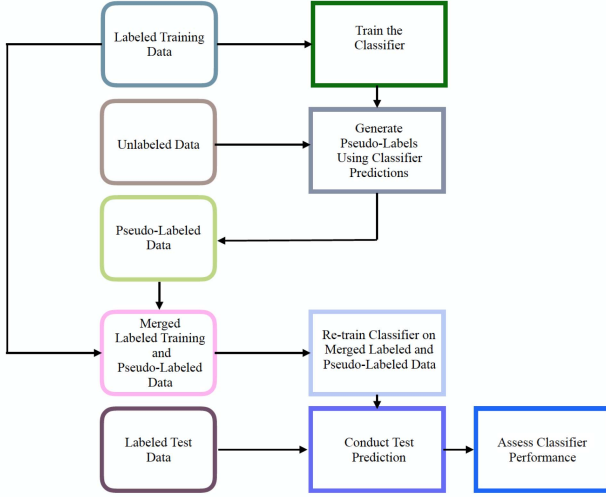


Fig. 1. Semi-supervised self-training framework for misbehavior detection.

updated based on the performance of the classifier at each iteration. The underlying assumption is that the classifier's F1-score serves as a reliable indicator of the quality of the pseudo-labels: a higher F1-score implies that the model is producing more trustworthy predictions, whereas a lower F1-score suggests increased uncertainty and the need for a stricter confidence requirement.

The adjustment mechanism is controlled by a variation value γ , which determines the magnitude of each update. If the classifier's F1-score exceeds the minimum acceptable value, the confidence threshold is decreased by γ to allow more samples to be incorporated into the self-training loop. Conversely, if the F1-score falls below the minimum acceptable value, the threshold is increased by γ to prevent the inclusion of low-confidence predictions. When the classifier's F1-score lies within an acceptable tolerance interval around the minimum acceptable value, the threshold remains unchanged. This tolerance interval, represented by δ , acts as a safety margin to avoid unstable or overly frequent updates caused by minor performance fluctuations.

Equation (1) formalizes the computation of the confidence threshold at iteration t_{i+1} :

$$C(t_{i+1}) = \begin{cases} C(t_i) + \gamma, & \text{if } b \leq a - \delta, \\ C(t_i), & \text{if } a - \delta < b < a + \delta, \\ C(t_i) - \gamma, & \text{if } b \geq a + \delta, \end{cases} \quad (1)$$

where $C(t_i)$ is the confidence threshold from the previous iteration, a denotes the minimum acceptable F1-score, b is the classifier's F1-score obtained in iteration i , and γ represents the update step size. The parameter δ defines the acceptable variation around a . In practice, the parameters a , γ , and δ are selected empirically through preliminary validation experiments: a is set according to the minimum performance level required for reliable pseudo-labelling, γ is chosen to ensure gradual and stable adaptation, and δ is tuned to balance update stability and responsiveness. The iterative labeling process stops when either all unlabeled samples have been assigned labels, no samples exceed the confidence threshold, or the maximum number of iterations is reached.

Algorithm 1 Adaptive Self-training Model

```

1: Input: Labeled set  $\{S_l\} = \{(x_i, y_i) \mid i = 1, 2, \dots, l\}$ 
2: Input: Unlabeled set  $\{S_u\} = \{x_j \mid j = l + 1, \dots, l + u\}$ 
3: Output: Updated labeled set  $\{S_l\}$ 
4: while  $S_u \neq \emptyset$  do
5:   Train classifier  $M$  using  $S_l$ 
6:   Predict pseudo-labels  $\hat{y}_j$  and confidence scores  $C(x_j)$  for all  $x_j \in S_u$ 
7:   Identify confidently labeled instances:
       
$$S_e = \{(x_j, \hat{y}_j) \mid C(x_j) \geq C_{\min}, x_j \in S_u\}$$

8:   Update labeled set:  $S_l \leftarrow S_l + S_e$ 
9:   Remove confident instances from unlabeled set:
       
$$S_u \leftarrow S_u - \{(x_j, \hat{y}_j) \in S_e\}$$

10: end while
11: Return: Final labeled set  $S_l$ 
  
```

Fig. 2. Adaptive self-training Algorithm.

Unlike the work in [21], we prioritize the F1-score as the pivotal metric among all performance evaluation metrics due to its ability to strike a balance between precision and recall. In the domain of misbehavior detection in vehicular networks, where the consequences of false positives and false negatives are significant, both precision and recall hold utmost importance. Therefore, the F1-score emerges as the most appropriate metric to assess the performance of previous iterations and fine-tune the confidence threshold accordingly. Fig. 2 shows the algorithm of the proposed adaptive self-training model.

V. PERFORMANCE EVALUATION AND RESULTS

This section presents the results of implementing adaptive confidence in the self-training method on the Vehicular Reference Misbehavior (VeReMi) dataset [22].

The VeReMi dataset is a publicly available benchmark designed for evaluating misbehavior detection mechanisms in cooperative intelligent transportation systems. The dataset synthesizes a variety of vehicular mobility traces and misbehavior scenarios under realistic communication conditions to reflect common anomalies and attacks in vehicular networks. It includes features such as position, speed, heading, and their associated noise measurements, along with annotated ground truth labels indicating normal or misbehavior events, making it particularly suitable for evaluating both traditional and machine learning-based intrusion detection models.

We systematically clean the dataset to ensure reliability and model readiness. Irrelevant columns, such as Unnamed: 0, are removed, and missing values are imputed. Numeric features are normalized, while categorical variables were encoded as needed, and duplicates are removed.

To address class imbalance, the Synthetic Minority Oversampling Technique (SMOTE) [23] is applied to the training set only, using the standard k -nearest neighbors configuration ($k = 5$). For each experiment, the dataset is shuffled (random seed = 42) and split into 70% training, 15% validation, and 15% testing subsets using stratified sampling. In self-training experiments, an unlabeled subset is constructed by removing labels from a randomly selected portion of the training data.

TABLE II
PERFORMANCE EVALUATION

95% of Data Initially Unlabeled				
Model	Accuracy	Precision	Recall	F1-Score
Semi-supervised (LR)	0.5453 $\pm$ 0.0105	0.8967 $\pm$ 0.0016	0.5361 $\pm$ 0.0157	0.6710 $\pm$ 0.0114
Semi-supervised (XGBoost)	0.7536 $\pm$ 0.0028	0.8872 $\pm$ 0.0008	0.8194 $\pm$ 0.0047	0.8521 $\pm$ 0.0026
Adaptive Semi-supervised (LR)	0.5522 $\pm$ 0.0115	0.8991 $\pm$ 0.0012	0.5434 $\pm$ 0.0160	0.6772 $\pm$ 0.0120
Adaptive Semi-supervised (XGBoost)	0.7552 $\pm$ 0.0035	0.8886 $\pm$ 0.0004	0.8197 $\pm$ 0.0047	0.8528 $\pm$ 0.0025
90% of Data Initially Unlabeled				
Model	Accuracy	Precision	Recall	F1-Score
Semi-supervised (LR)	0.5451 $\pm$ 0.0158	0.8990 $\pm$ 0.0009	0.5342 $\pm$ 0.0205	0.6701 $\pm$ 0.0158
Semi-supervised (XGBoost)	0.7576 $\pm$ 0.0057	0.8845 $\pm$ 0.0022	0.8278 $\pm$ 0.0064	0.8552 $\pm$ 0.0037
Adaptive Semi-supervised (LR)	0.5555 $\pm$ 0.0160	0.8974 $\pm$ 0.0007	0.5489 $\pm$ 0.0208	0.6809 $\pm$ 0.0161
Adaptive Semi-supervised (XGBoost)	0.7639 $\pm$ 0.0056	0.8871 $\pm$ 0.0018	0.8331 $\pm$ 0.0060	0.8592 $\pm$ 0.0037
85% of Data Initially Unlabeled				
Model	Accuracy	Precision	Recall	F1-Score
Semi-supervised (LR)	0.5396 $\pm$ 0.0164	0.8999 $\pm$ 0.0008	0.5263 $\pm$ 0.0221	0.6642 $\pm$ 0.0171
Semi-supervised (XGBoost)	0.7758 $\pm$ 0.0038	0.8836 $\pm$ 0.0015	0.8533 $\pm$ 0.0054	0.8681 $\pm$ 0.0026
Adaptive Semi-supervised (LR)	0.5563 $\pm$ 0.0168	0.8963 $\pm$ 0.0005	0.5508 $\pm$ 0.0222	0.6821 $\pm$ 0.0171
Adaptive Semi-supervised (XGBoost)	0.7846 $\pm$ 0.0207	0.8828 $\pm$ 0.0025	0.8660 $\pm$ 0.0305	0.8740 $\pm$ 0.0141
80% of Data Initially Unlabeled				
Model	Accuracy	Precision	Recall	F1-Score
Semi-supervised (LR)	0.5453 $\pm$ 0.0105	0.8967 $\pm$ 0.0016	0.5361 $\pm$ 0.0157	0.6710 $\pm$ 0.0114
Semi-supervised (XGBoost)	0.7536 $\pm$ 0.0028	0.8872 $\pm$ 0.0008	0.8194 $\pm$ 0.0047	0.8521 $\pm$ 0.0026
Adaptive Semi-supervised (LR)	0.5522 $\pm$ 0.0115	0.8991 $\pm$ 0.0012	0.5434 $\pm$ 0.0160	0.6772 $\pm$ 0.0120
Adaptive Semi-supervised (XGBoost)	0.7552 $\pm$ 0.0035	0.8886 $\pm$ 0.0004	0.8197 $\pm$ 0.0047	0.8528 $\pm$ 0.0025
75% of Data Initially Unlabeled				
Model	Accuracy	Precision	Recall	F1-Score
Semi-supervised (LR)	0.5329 $\pm$ 0.0158	0.8956 $\pm$ 0.0034	0.5207 $\pm$ 0.0185	0.6585 $\pm$ 0.0153
Semi-supervised (XGBoost)	0.7983 $\pm$ 0.0026	0.8784 $\pm$ 0.0015	0.8901 $\pm$ 0.0047	0.8842 $\pm$ 0.0018
Adaptive Semi-supervised (LR)	0.5646 $\pm$ 0.0160	0.8932 $\pm$ 0.0023	0.5640 $\pm$ 0.0192	0.6913 $\pm$ 0.0152
Adaptive Semi-supervised (XGBoost)	0.8415 $\pm$ 0.0039	0.8743 $\pm$ 0.0015	0.9539 $\pm$ 0.0072	0.9123 $\pm$ 0.0026

To ensure reproducibility, all models are trained using fixed random seeds. All experiments are repeated over five different shuffle seeds, and final performance is reported as the average and standard deviation of these runs. Evaluation metrics include precision, recall, F1-score, accuracy, and confusion matrices. The exact hyperparameter configurations and software environment (Python version and library versions) are documented in this paper to support reproducibility.

A. Classification Performance

We employ two different classifiers to obtain our results: Logistic Regression (LR) and Extreme Gradient Boosting (XGBoost). Table II presents the experimental results for initial unlabeled data percentages of 95%, 90%, 85%, 80%, and 75%. Each result reflects the average of 5 independent runs, with standard deviation reported accordingly.

Across all labeling scenarios, the Adaptive Semi-supervised (XGBoost) model consistently achieves the highest accuracy. At 95% unlabeled data, it attains the best performance (0.7552 $\pm$ 0.0035), slightly outperforming the standard Semi-supervised (XGBoost) model. This superiority remains consistent at 90% (0.7639 $\pm$ 0.0056) and 85% (0.7846 $\pm$ 0.0207), and becomes most pronounced at 75% unlabeled data, where it reaches 0.8415 $\pm$ 0.0039. In contrast, Logistic Regression (LR)-based methods remain substantially lower (approximately 0.53–0.56) across all settings. Therefore, Adaptive Semi-supervised (XGBoost) shows the best overall accuracy, with performance improving as the proportion of labeled data increases.

Precision values are highest for LR-based models across all labeling ratios. In particular, Adaptive Semi-supervised (LR) achieves the highest precision in most cases (e.g., 0.8991 $\pm$ 0.0012 at 95%, 0.8990 $\pm$ 0.0009 at 90%, and 0.8963 $\pm$ 0.0005 at 85%), while Semi-supervised (LR) slightly leads at 75% (0.8956 $\pm$ 0.0034). XGBoost-based models maintain slightly lower but stable precision values (approximately 0.87–0.89). Hence, LR-based approaches yield the best overall precision, indicating fewer false positives, albeit at the expense of recall.

Recall is consistently highest for Adaptive Semi-supervised (XGBoost) across all scenarios. At 95% unlabeled data, it achieves 0.8197 $\pm$ 0.0047, marginally exceeding the standard XGBoost variant. As the proportion of labeled data increases, recall improves steadily, reaching 0.8660 $\pm$ 0.0305 at 85% and peaking at 0.9539 $\pm$ 0.0072 at 75% unlabeled data. In contrast, LR-based models remain limited to approximately 0.52–0.56 recall across all settings. Thus, Adaptive Semi-supervised (XGBoost) demonstrates the best recall performance, particularly as more labeled data become available.

The F1-score, which balances precision and recall, is consistently highest for Adaptive Semi-supervised (XGBoost) under every labeling ratio. It achieves 0.8528 $\pm$ 0.0025 (95%), 0.8592 $\pm$ 0.0037 (90%), 0.8740 $\pm$ 0.0141 (85%), and reaches a maximum of 0.9123 $\pm$ 0.0026 at 75% unlabeled data. The standard Semi-supervised (XGBoost) ranks second in all cases, while both LR-based methods remain substantially lower (approximately 0.66–0.69). Therefore, Adaptive Semi-supervised (XGBoost) provides the best overall balance between precision and recall.

Metric-wise comparison indicates that Adaptive Semi-supervised (XGBoost) achieves the best performance in terms of Accuracy, Recall, and F1-score across all labeling ratios, while LR-based models yield the highest Precision. Considering balanced classification performance (F1-score), Adaptive Semi-supervised (XGBoost) is the overall best-performing method.

Beyond accurate detection, computational efficiency is a critical requirement for real-time misbehavior detection in vehicular networks. The proposed model is trained offline and subsequently deployed at roadside units (RSUs) for online inference, thereby minimizing runtime overhead during operation. The inference phase relies on lightweight feed-forward computations, which enables timely decision-making and supports real-time deployment. Although the training process incurs higher computational cost, it is conducted offline and therefore does not affect the responsiveness of the operational system.

B. Hyperparameter Optimization

To optimize the performance of the Adaptive Semi-supervised XGBoost model, we employ a computationally efficient hyperparameter tuning strategy based on randomized search [24] combined with validation-driven self-training. The hyperparameter optimization focuses on the most influential parameters of XGBoost, including maximum tree depth (max_depth), learning rate (η), minimum child weight (min_child_weight), subsample ratio ($subsample$), column sampling ratio ($colsample_bytree$), L2 regularization (λ), and L1 regularization (α). The objective is to maximize the F1-score on a validation set derived from the labeled portion of the training data. Hyperparameter optimization is performed prior to the self-training iterations using labeled validation data.

Formally, let $\mathcal{H}$ denote the hyperparameter search space, defined as:

$$\mathcal{H} = \begin{cases} max_depth \in \{3, 4, 5\}, \\ \eta \in [0.03, 0.1], \\ min_child_weight \in \{1, 3, 5\}, \\ subsample \in [0.7, 0.9], \\ colsample_bytree \in [0.7, 0.9], \\ \lambda \in [1, 10], \\ \alpha \in [0, 0.5] \end{cases}$$

where parameters enclosed in curly braces $\{\cdot\}$ are selected from discrete sets, and parameters in brackets $[\cdot]$ are treated as continuous ranges sampled uniformly during randomized search.

The tuning procedure solves the following optimization problem:

$$\theta^* = \arg \max_{\theta \in \mathcal{H}} F1(\theta; X_{val}, y_{val}),$$

where θ represents a candidate hyperparameter configuration, and $F1(\cdot)$ is the F1-score evaluated on the validation set (X_{val}, y_{val}) .

To improve computational efficiency, we use several strategies during hyperparameter tuning. First, only a subset of $n_{iter} = 10$ configurations is sampled from the hyperparameter space $\mathcal{H}$ using randomized search, avoiding the computational burden of

exhaustive grid search. Second, the number of boosting rounds ($n_{estimators}$) is temporarily reduced to 120 during the tuning phase to accelerate evaluation. Third, self-training iterations incorporate early stopping: the process terminates when the F1-score on the validation set fails to improve beyond a small threshold (δ), thereby preventing unnecessary computations. Finally, the optimal hyperparameters obtained for higher unlabeled data ratios are used as starting points for lower unlabeled ratios, further reducing the number of tuning iterations while maintaining model performance.

Once the optimal hyperparameters θ^* are identified, they are used to train the final Adaptive Semi-supervised XGBoost model on the labeled and pseudo-labeled data, and evaluated on the test set. This procedure ensures a balance between computational efficiency and model performance while maintaining a fair and reproducible comparison across different labeling scenarios.

Table III presents the performance of the proposed Adaptive Semi-supervised XGBoost model before and after applying hyperparameter optimization across different fractions of initially unlabeled data. Across all unlabeled data ratios, the optimized model consistently shows improved performance, as indicated by higher mean values in accuracy, precision, recall, and F1-score. For example, when 75% of the data is initially unlabeled, the accuracy improves from 0.8415 ± 0.0039 to 0.8681 ± 0.0028 , while the F1-score increases from 0.9123 ± 0.0026 to 0.9403 ± 0.0031 , reflecting a substantial gain in the balance between precision and recall. The improvements are also observed for higher unlabeled fractions, although the magnitude is smaller, at 95% unlabeled, accuracy rises modestly from 0.7552 ± 0.0035 to 0.7621 ± 0.0013 , and F1-score from 0.8528 ± 0.0025 to 0.8585 ± 0.0021 . Importantly, the relatively narrow 95% confidence intervals indicate that the observed improvements are statistically meaningful and unlikely to be due to random variation in the data splits. These results show that hyperparameter optimization not only enhances overall predictive performance but also stabilizes the model, particularly under conditions of limited labeled data, highlighting the effectiveness of the proposed adaptive XGBoost framework in semi-supervised scenarios.

Fig. 3 shows the percentage improvement in accuracy, precision, recall, and F1-score after applying hyperparameter optimization to the proposed Adaptive semi-supervised XGBoost model across different fractions of initially unlabeled data. The figure clearly illustrates that optimization consistently enhances model performance, with the most substantial gains observed when 75% of the data is initially unlabeled, particularly in recall and F1-score, highlighting the effectiveness of the optimization in improving both the sensitivity and overall balance of the model under semi-supervised scenarios.

C. Comparison With Previous Study

To the best of our knowledge, limited research has focused on misbehavior detection in inter-vehicle communications using semi-supervised learning techniques. To contextualize our contribution, we compare our method with one of the most recent semi-supervised misbehavior detection approaches for

TABLE III
PERFORMANCE EVALUATION AFTER APPLYING HYPERPARAMETER OPTIMIZATION FOR THE PROPOSED ADAPTIVE SEMI-SUPERVISED XGBOOST METHOD

95% of Data Initially Unlabeled				
	Accuracy	Precision	Recall	F1-Score
Before	0.7552 ± 0.0035	0.8886 ± 0.0004	0.8197 ± 0.0047	0.8528 ± 0.0025
After	0.7621 ± 0.0013	0.8942 ± 0.0018	0.8256 ± 0.0021	0.8585 ± 0.0021
90% of Data Initially Unlabeled				
	Accuracy	Precision	Recall	F1-Score
Before	0.7639 ± 0.0056	0.8871 ± 0.0018	0.8331 ± 0.0060	0.8592 ± 0.0037
After	0.7714 ± 0.0028	0.8947 ± 0.0034	0.8411 ± 0.0024	0.8671 ± 0.0035
85% of Data Initially Unlabeled				
	Accuracy	Precision	Recall	F1-Score
Before	0.7846 ± 0.0207	0.8828 ± 0.0025	0.8660 ± 0.0305	0.8740 ± 0.0141
After	0.7893 ± 0.0054	0.8939 ± 0.0024	0.8673 ± 0.0043	0.8804 ± 0.0024
80% of Data Initially Unlabeled				
	Accuracy	Precision	Recall	F1-Score
Before	0.7552 ± 0.0035	0.8886 ± 0.0004	0.8197 ± 0.0047	0.8528 ± 0.0025
After	0.7916 ± 0.0026	0.8961 ± 0.0026	0.8218 ± 0.00032	0.8573 ± 0.0036
75% of Data Initially Unlabeled				
	Accuracy	Precision	Recall	F1-Score
Before	0.8415 ± 0.0039	0.8743 ± 0.0015	0.9539 ± 0.0072	0.9123 ± 0.0026
After	0.8681 ± 0.0028	0.9093 ± 0.0025	0.9735 ± 0.0028	0.9403 ± 0.0031

TABLE IV
COMPARISON OF PERFORMANCE WITH THE PREVIOUS SEMI-SUPERVISED VEHICULAR MISBEHAVIOR DETECTION MODEL

	Accuracy	Precision	Recall	F1-Score
SVMDformer [13]	0.8341 ± 0.0012	0.8245 ± 0.0031	0.8673 ± 0.0031	0.8454 ± 0.0028
Adaptive Semi-supervised (XGBoost)	0.8681 ± 0.0028	0.9093 ± 0.0025	0.9735 ± 0.0028	0.9403 ± 0.0031

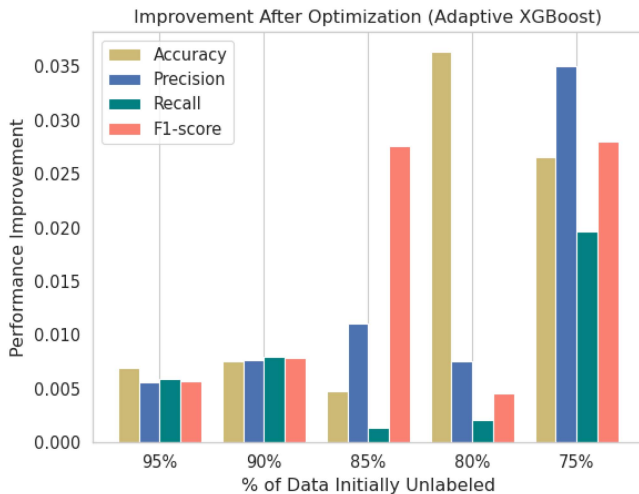


Fig. 3. Performance improvement after applying hyperparameter optimization.

vehicular networks, namely the threshold algorithm of SVMDformer [13], as summarized in Table IV. Although SVMDformer was originally evaluated on the VeReMi dataset, we re-assess

its thresholding mechanism under the same conditions on the portion of the dataset used in our experiments, enabling a fair and direct comparison with our proposed self-training misbehavior detection model with adaptive confidence thresholding.

When comparing SVMDformer and the proposed Adaptive Semi-supervised XGBoost model on the 75% initially unlabeled dataset, the latter shows superior performance across all metrics. While SVMDformer achieves an accuracy of 0.8341 ± 0.0012 , the Adaptive XGBoost improves this to 0.8681 ± 0.0028 , indicating a more reliable overall prediction. Precision increases substantially from 0.8245 ± 0.0031 to 0.9093 ± 0.0025 , reflecting a significant reduction in false positives. The most notable difference is observed in recall, which rises from 0.8673 ± 0.0031 for SVMDformer to 0.9735 ± 0.0028 for Adaptive XGBoost, highlighting the model's ability to detect nearly all misbehavior instances. Consequently, the F1-score improves from 0.8454 ± 0.0028 to 0.9403 ± 0.0031 , demonstrating a stronger balance between precision and recall.

The performance gap between SVMDformer and our Adaptive Semi-supervised XGBoost arises from their thresholding strategies. SVMDformer prioritizes recall by setting thresholds based only on true positives and false negatives, which can lead to overly permissive decisions. In

contrast, our method adaptively adjusts thresholds based on the previous iteration's F1-score, balancing precision and recall, preventing low-quality pseudo-label propagation, and stabilizing learning. This adaptive framework, combined with gradient-boosted learning, provides a more robust and reliable solution for detecting misbehavior in vehicular networks.

From a computational perspective, the overall complexity of the proposed framework is primarily governed by the base learner, namely XGBoost. For a dataset with n samples and d features, and using T boosting iterations (trees), the training complexity of XGBoost can be approximated as $\mathcal{O}(T \cdot n \cdot \log n \cdot d)$, depending on the specific tree construction strategy [25], [26]. In contrast, transformer-based models such as SVMdformer rely on self-attention mechanisms, whose per-layer computational cost scales as $\mathcal{O}(n^2 \cdot d + n \cdot d^2)$, where n denotes the input sequence length and d the embedding dimension [27], [28]. It should be noted that these complexities are not directly comparable due to fundamental differences in data representation and model architecture (i.e., tabular tree-based learning versus sequence-based attention models). The proposed adaptive thresholding mechanism introduces only a negligible additional cost, as it involves simple scalar updates based on the F1-score at each iteration and does not affect the dominant training complexity of the underlying model.

VI. CONCLUSION

In this paper, we introduced an adaptive misbehavior detection framework for VANETs based on semi-supervised self-training. Using the VeReMi dataset, the proposed model was evaluated across varying unlabeled data scenarios using accuracy, precision, recall, and F1-score as performance metrics, and compared against existing approaches. The experimental results showed the effectiveness of the adaptive learning strategy in detecting malicious vehicular behavior. After applying hyperparameter optimization, the performance of the adaptive semi-supervised XGBoost model improved consistently across all evaluation metrics. Moreover, when evaluated under identical conditions, the proposed approach surpassed the performance of the SVMdformer model [13]. These findings highlight the robustness of the proposed adaptive self-training framework, particularly in large-scale vehicular environments with limited labeled data. By integrating adaptive thresholding with optimized gradient boosting, the proposed method advances the effectiveness of semi-supervised misbehavior detection in VANETs.

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Francois Chan (Senior Member, IEEE) received the BEng degree in electrical engineering from McGill University, Montreal, Canada, and the MScA and PhD degrees in electrical engineering from Polytechnique Montréal, Canada. He is currently a professor with the Department of Electrical and Computer Engineering, Royal Military College of Canada, Kingston, ON, Canada. He is also an adjunct professor and a visiting professor with the University of Ottawa. He was a visiting researcher with the University of California, Irvine, in Fall 2002 and Fall 2005. His current research interests include machine learning, digital communications, wireless communications, and digital signal processing.



Elnaz Limouchi (Member, IEEE) received the PhD degree in electrical engineering from the Department of Computer and Electrical Engineering and Computer Science, Florida Atlantic University, Boca Raton, FL, USA, in 2017. Her research interests include vehicular networking, artificial intelligence, and machine learning applications in wireless networks. She is a Research Associate collaborating with the Royal Military College of Canada, focusing on data-driven and intelligent techniques for wireless and networked systems. She was a recipient of the Jaime A. Borras

Family Endowed Scholarship, the Florida Atlantic University Women in Engineering and Computer Science Award for outstanding academic excellence, the FAU Graduate Research Day Award, and the ECEdHA Professional Development Award from the National Science Foundation.